INHERITANCE

Real-Time Python Assignments • Classroom Practice

Think in REAL applications!

Instagram • Netflix • Amazon • Banking

AWS • Kubernetes • Cybersecurity • E-commerce

Build the classes, run the code, and explain the relationship.

MEMORY TRICK

Single = One parent -> One child
Multiple = Multiple parents -> One child
Multilevel = Parent -> Child -> Grandchild
Hierarchical = One parent -> Multiple children

Let's code!

1. SINGLE INHERITANCE

One parent -> One child

1. *Instagram: Create Post with upload() and like(). Create Reel that inherits Post and adds add_music(). Create a Reel object and execute all three methods.*

2. *Netflix: Create NetflixUser with login() and watch_movie(). Create PremiumUser that inherits it and adds download_movie().*

3. *Amazon: Create AmazonUser with login() and view_orders(). Create PrimeUser that inherits it and adds prime_delivery().*

4. *Banking: Create BankAccount with login() and check_balance(). Create SavingsAccount that inherits it and adds calculate_interest().*

5. *AWS: Create CloudUser with login() and view_dashboard(). Create AWSAdmin that inherits it and adds create_iam_user().*

2. MULTIPLE INHERITANCE

Multiple parents -> One child

6. *Influencer: Create ContentCreator with create_content() and BusinessOwner with sell_product(). Create Influencer that inherits from both and execute both methods.*

7. *SDET: Create Developer with write_code() and Tester with test_application(). Create SDET inheriting from both and demonstrate both capabilities.*

8. *DevSecOps: Create Developer with deploy_code() and SecurityEngineer with scan_code(). Create DevSecOpsEngineer inheriting from both.*

9. *Cloud Security: Create CloudEngineer with create_infrastructure() and SecurityEngineer with secure_infrastructure(). Create CloudSecurityEngineer inheriting from both.*

10. *E-commerce: Create SalesEmployee with sell_product() and SupportEmployee with handle_customer(). Create EcommerceEmployee inheriting from both.*

3. MULTILEVEL INHERITANCE

Parent -> Child -> Grandchild

11 *Banking: Create BankAccount -> CorporateAccount -> CorporateAdmin. Give each level at least one unique method and execute all methods using CorporateAdmin.*

12 *Mobile: Create Phone -> SmartPhone -> IPhone. Phone has call(), SmartPhone has browse_internet(), and IPhone has face_id().*

13 *Cloud Security: Create CloudUser -> CloudEngineer -> CloudSecurityEngineer. Give each class one unique cloud/security method and execute all from the final child.*

14 *Kubernetes: Create KubernetesUser -> KubernetesDeveloper -> KubernetesAdmin. Add realistic methods at each level and demonstrate all inherited methods.*

15 *Employee Management: Create Employee -> Developer -> SeniorDeveloper. Give each class different responsibilities and execute all methods using SeniorDeveloper.*

4. HIERARCHICAL INHERITANCE

One parent -> Multiple children

16 *YouTube: Create YouTubeUser with watch(). Create StudentUser and GamerUser as children, each with its own additional functionality.*

17 *Cloud Platform: Create CloudUser with login(). Create Admin, Developer, and Auditor as children with separate real-world responsibilities.*

18 *Banking: Create BankAccount with common functionality. Create SavingsAccount and CurrentAccount as child classes with different functionality.*

19 *Food Delivery: Create FoodDeliveryUser with login(). Create Customer and Restaurant as children with their own operations.*

20 *Cybersecurity: Create SecurityUser with login(). Create SOCAnalyst and SecurityAdmin as children with different security responsibilities.*

5. FINAL CHALLENGE

Real-Time Security Platform

Question 21 - Design the system

Create this inheritance chain:
SecurityUser -> CloudSecurityUser -> CloudSecurityAdmin

Add realistic methods at each level.
Create a CloudSecurityAdmin object and execute methods from ALL THREE levels.

STUDENT CHECKLIST

- ☐ *Parent class created*
- ☐ *Child class inherits correctly*
- ☐ *Inherited methods reused*
- ☐ *Child-specific methods added*
- ☐ *IS-A relationship explained*